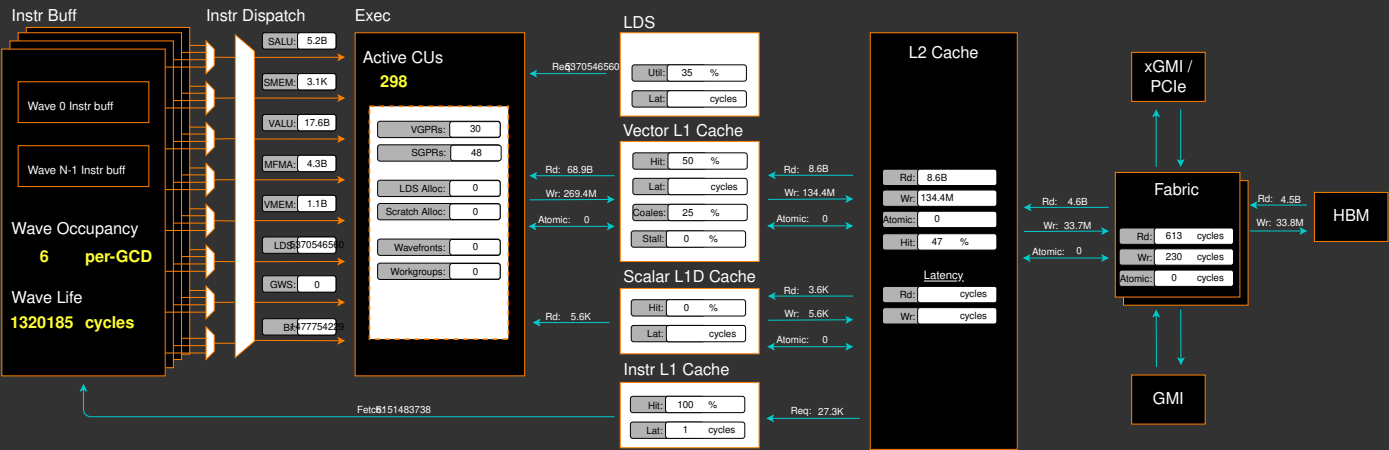
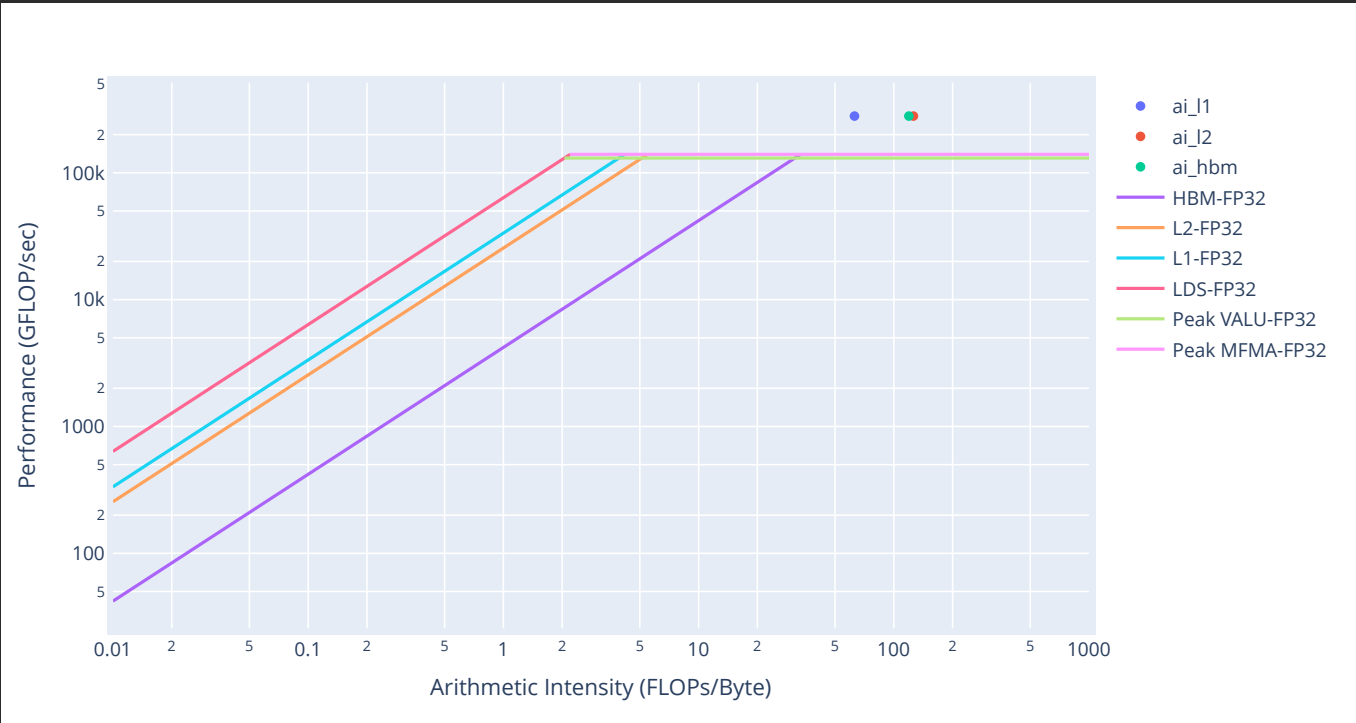


Placeholder. Guided Analysis coming soon...



Empirical Roofline Analysis (Flops)



2. System Speed-of-Light

2.1 Speed-of-Light

Metric	Avg	Unit	Peak	Pct of Peak
VALU FLOPs	4.00	Gflop/s	81715.20	0.00
VALU IOPs	1710.82	Giop/s	81715.20	2.09
MFMA FLOPs (F8)	0.00	Gflop/s	2614886.40	0.00
MFMA FLOPs (BF16)	0.00	Gflop/s	1307443.20	0.00
MFMA FLOPs (F16)	280163.54	Gflop/s	1307443.20	21.43
MFMA FLOPs (F32)	0.00	Gflop/s	163430.40	0.00
MFMA FLOPs (F64)	0.00	Gflop/s	163430.40	0.00
MFMA FLOPs (F6F4)		Gflop/s		
MFMA IOPs (Int8)	0.00	Giop/s	2614886.40	0.00
Active CUs	298.00	Cus	304.00	98.03
SALU Utilization	6.04	Pct	100.00	6.04
VALU Utilization	20.31	Pct	100.00	20.31
MFMA Utilization	39.14	Pct	100.00	39.14
VMEM Utilization	1.23	Pct	100.00	1.23
Branch Utilization	2.17	Pct	100.00	2.17
VALU Active Threads	64.00	Threads	64.00	100.00
IPC	0.39	Instr/cycle	5.00	7.72
Wavefront Occupancy	1777.51	Wavefronts	9728.00	18.27
Theoretical LDS Bandwidth	15275.08	Gb/s	81715.20	18.69
LDS Bank Conflicts/Access	0.00	Conflicts/access	32.00	0.00
vL1D Cache Hit Rate	50.09	Pct	100.00	50.09
vL1D Cache BW	8909.12	Gb/s	81715.20	10.90
L2 Cache Hit Rate	47.40	Pct	100.00	47.40
L2 Cache BW	4446.19	Gb/s	34406.40	12.92
L2-Fabric Read BW	2334.46	Gb/s	5324.80	43.84
L2-Fabric Write BW	8.58	Gb/s	5324.80	0.16
L2-Fabric Read Latency	613.30	Cycles		
L2-Fabric Write Latency	230.09	Cycles		
sL1D Cache Hit Rate	0.00	Pct	100.00	0.00
sL1D Cache BW	0.00	Gb/s	21504.00	0.00
L1I Hit Rate	100.00	Pct	100.00	100.00
L1I BW	1562.78	Gb/s	21504.00	7.27
L1I Fetch Latency		Cycles		

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric	Avg	Min	Max	Unit
CPF Utilization	100.00	100.00	100.00	Pct
CPF Stall	0.00	0.00	0.00	Pct
CPF-L2 Utilization	0.03	0.00	0.23	Pct
CPF-L2 Stall	0.00	0.00	0.00	Pct
CPF-UTCL1 Stall	0.00	0.00	0.00	Pct

5.2 Packet Processor

Metric	Avg	Min	Max	Unit
CPC SYNC FIFO Full Rate				Pct
CPC CANE Stall Rate				Pct
CPC ADC Utilization				Pct
CPC Utilization	99.90	98.41	100.00	Pct
CPC Stall Rate	0.42	0.10	0.56	Pct
CPC Packet Decoding Utilization	2.30	0.44	3.45	Pct
CPC-Workgroup Manager Utilization	97.65	91.83	102.00	Pct
CPC-L2 Utilization	0.00	0.00	0.00	Pct
CPC-UTCL1 Stall	0.00	0.00	0.00	Pct
CPC-UTCL2 Utilization	0.00	0.00	0.00	Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric	Avg	Min	Max	Unit
Schedule-Pipe Wave Occupancy				Wave
Accelerator Utilization	100.17	97.52	102.83	Pct
Scheduler-Pipe Utilization	0.00	0.00	0.00	Pct
Scheduler-Pipe Wave Utilization				Pct
Workgroup Manager Utilization	97.93	95.60	99.99	Pct
Shader Engine Utilization	97.94	95.43	104.19	Pct
SIMD Utilization	97.53	94.59	99.90	Pct
Dispatched Workgroups	0.00	0.00	0.00	Workgroups
Dispatched Wavefronts	0.00	0.00	0.00	Wavefronts
VGPR Writes				Cycles/wave
SGPR Writes				Cycles/wave

6.2 Workgroup Manager - Resource Allocation

Metric	Avg	Min	Max	Unit
Not-scheduled Rate (Workgroup Manager)	0.00	0.00	0.00	Pct
Not-scheduled Rate (Scheduler-Pipe)	0.00	0.00	0.00	Pct
Scheduler-Pipe FIFO Full Rate				Pct
Scheduler-Pipe Stall Rate	0.00	0.00	0.00	Pct
Scratch Stall Rate	0.00	0.00	0.00	Pct
Insufficient SIMD Waveslots	0.00	0.00	0.00	Pct
Insufficient SIMD VGPRs	0.00	0.00	0.00	Pct
Insufficient SIMD SGPRs	0.00	0.00	0.00	Pct
Insufficient CU LDS	0.00	0.00	0.00	Pct
Insufficient CU Barriers	0.00	0.00	0.00	Pct
Reached CU Workgroup Limit	0.00	0.00	0.00	Pct
Reached CU Wavefront Limit	0.00	0.00	0.00	Pct

7. Wavefront

7.1 Wavefront Launch Stats

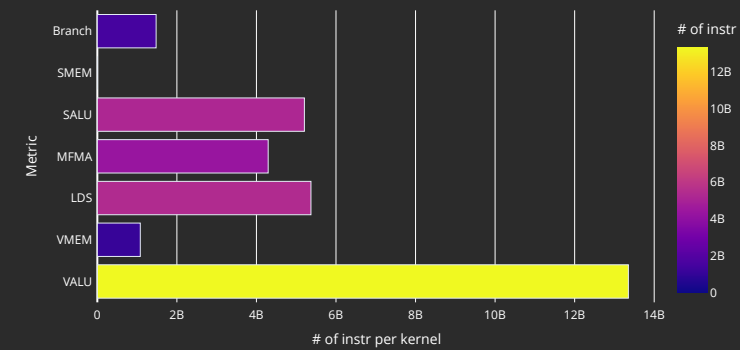
↕Metric	↕Avg	↕Min	↕Max	↕Unit
Grid Size	25165824.00	16777216.00	33554432.00	Work items
Workgroup Size	384.00	256.00	512.00	Work items
Total Wavefronts	0.00	0.00	0.00	Wavefronts
Saved Wavefronts	228.00	0.00	2432.00	Wavefronts
Restored Wavefronts	218.63	0.00	2282.00	Wavefronts
VGPRs	30.00	20.00	40.00	Registers
AGPRs	162.00	132.00	192.00	Registers
SGPRs	48.00	48.00	48.00	Registers
LDS Allocation	0.00	0.00	0.00	Bytes
Scratch Allocation	0.00	0.00	0.00	Bytes/workitem

7.2 Wavefront Runtime Stats

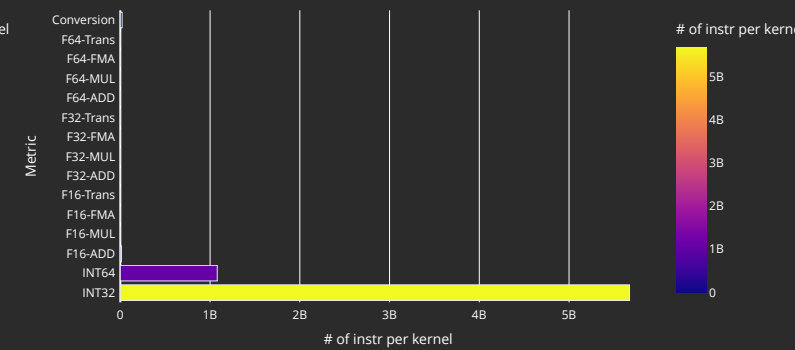
↕Metric	↕Avg	↕Min	↕Max	↕Unit
Kernel Time (Nanosec)	251305260.30	241297375.75	258926448.25	Ns
Kernel Time (Cycles)	292479100.07	261682905.63	392016010.88	Cycle
Instructions per wavefront	89036.27	70301.22	107540.81	Instr/wavefront
Wave Cycles	501487515003.25	365763059808.00	622468643880.00	Cycles per kernel
Dependency Wait Cycles	141746631770.75	89130480040.00	194457533040.00	Cycles per kernel
Issue Wait Cycles	218745137356.75	152184059316.00	285530073532.00	Cycles per kernel
Active Cycles	142610345020.25	123753912032.00	161419120480.00	Cycles per kernel
Wavefront Occupancy	1777.51	1184.23	2371.38	Wavefronts

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix



10.2 VALU Arithmetic Instr Mix



10.3 VMEM Instr Mix

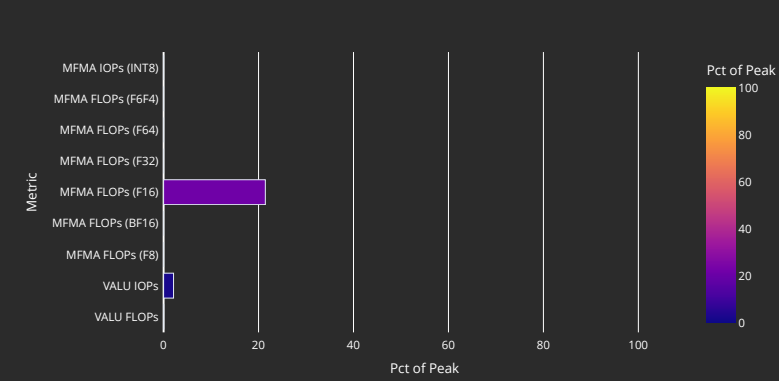
Metric	Avg	Min	Max	Unit
Global/Generic Instr	1077936128.00	1077936128.00	1077936128.00	Instr per kernel
Global/Generic Read	1073741824.00	1073741824.00	1073741824.00	Instr per kernel
Global/Generic Write	4194304.00	4194304.00	4194304.00	Instr per kernel
Global/Generic Atomic	0.00	0.00	0.00	Instr per kernel
Spill/Stack Instr	2191870.00	2097152.00	2931680.00	Instr per kernel
Spill/Stack Coalesceable Instr				Instr per kernel
Spill/Stack Read	2146915.00	2097152.00	2533424.00	Instr per kernel
Spill/Stack Write	20064.00	0.00	321024.00	Instr per kernel
Spill/Stack Atomic	0.00	0.00	0.00	Instr per kernel

10.4 MFMA Arithmetic Instr Mix

Metric	Avg	Min	Max	Unit
MFMA-I8	0.00	0.00	0.00	Instr per kernel
MFMA-F8	0.00	0.00	0.00	Instr per kernel
MFMA-F16	4294967296.00	4294967296.00	4294967296.00	Instr per kernel
MFMA-BF16	0.00	0.00	0.00	Instr per kernel
MFMA-F32	0.00	0.00	0.00	Instr per kernel
MFMA-F64	0.00	0.00	0.00	Instr per kernel
MFMA-F6F4				Instr per kernel

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light



11.2 Pipeline Stats

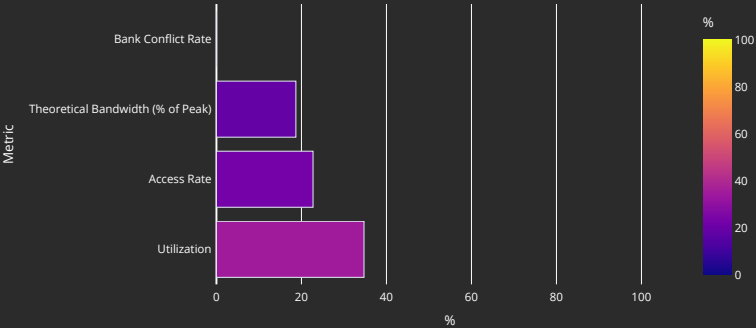
Metric	Avg	Min	Max	Unit
IPC	0.39	0.24	0.48	Instr/cycle
IPC (Issued)	0.86	0.84	0.88	Instr/cycle
SALU Utilization	6.04	3.51	7.82	Pct
VALU Utilization	20.31	13.26	24.51	Pct
VALU Co-Issue Efficiency				Pct
VMEM Utilization	1.23	0.91	1.36	Pct
Branch Utilization	2.17	1.35	2.70	Pct
VALU Active Threads	64.00	63.99	64.00	Threads
MFMA Utilization	39.14	28.83	43.19	Pct
MFMA Instr Cycles	32.00	32.00	32.00	Cycles/instr
VMEM Latency				Cycles
SMEM Latency				Cycles

11.3 Arithmetic Operations

Metric	Avg	Min	Max	Unit
FLOPs (Total)	70369750810624.00	70369683701760.00	70369817919488.00	Ops per kernel
IOPs (Total)	432256896576.00	311972331520.00	552545870848.00	Ops per kernel
F8 OPs	0.00	0.00	0.00	Ops per kernel
F16 OPs	70369549484032.00	70369549484032.00	70369549484032.00	Ops per kernel
BF16 OPs	0.00	0.00	0.00	Ops per kernel
F32 OPs	201326592.00	134217728.00	268435456.00	Ops per kernel
F64 OPs	0.00	0.00	0.00	Ops per kernel
F6F4 OPs				Ops per kernel
INT8 OPs	0.00	0.00	0.00	Ops per kernel

12. Local Data Share (LDS)

12.1 Speed-of-Light

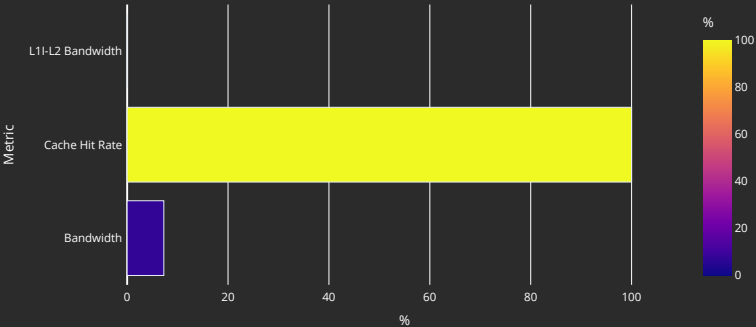


12.2 LDS Stats

Metric	Avg	Min	Max	Unit
LDS Instrs	5370546560.00	3761766400.00	6979321856.00	Instr per kernel
LDS LOAD				Instr per kernel
LDS STORE				Instr per kernel
LDS ATOMIC				Instr per kernel
LDS LOAD Bandwidth				Gbps
LDS STORE Bandwidth				Gbps
LDS ATOMIC Bandwidth				Gbps
Theoretical Bandwidth	3848293187584.00	3298534883328.00	4398066434048.00	Bytes per kernel
LDS Latency				Cycles
Bank Conflicts/Access	0.00	0.00	0.00	Conflicts/access
Index Accesses	30064790528.00	25769803776.00	34359894016.00	Cycles per kernel
Atomic Return Cycles	0.00	0.00	0.00	Cycles per kernel
Bank Conflict	0.00	0.00	0.00	Cycles per kernel
Addr Conflict	0.00	0.00	0.00	Cycles per kernel
Unaligned Stall	0.00	0.00	0.00	Cycles per kernel
Mem Violations	0.00	0.00	0.00	Accesses per kernel
LDS Command FIFO Full Rate				Cycles per kernel
LDS Data FIFO Full Rate				Cycles per kernel

13. Instruction Cache

13.1 Speed-of-Light



13.2 Instruction Cache Accesses

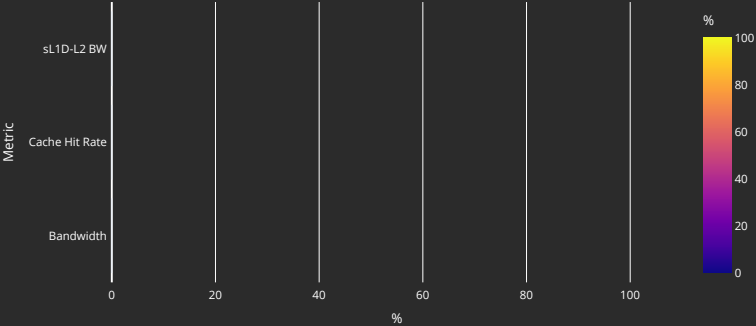
Metric	Avg	Min	Max	Unit
Req	6151483737.56	5287182336.00	7016808202.00	Req per kernel
Hits	6151415008.69	5287178797.00	7016746468.00	Hits per kernel
Misses - Non Duplicated	710.88	320.00	2530.00	Misses per kernel
Misses - Duplicated	4275.13	2677.00	17094.00	Misses per kernel
Cache Hit Rate	100.00	100.00	100.00	Pct
Instruction Fetch Latency				Cycles

13.3 Instruction Cache - L2 Interface

Metric	Avg	Min	Max	Unit
L1I-L2 Bandwidth	1746092.00	921600.00	5099520.00	Bytes per kernel

14. Scalar L1 Data Cache

14.1 Speed-of-Light



14.2 Scalar L1D Cache Accesses

Metric	Avg	Min	Max	Unit
Req	5599.06	0.00	89585.00	Req per kernel
Hits	0.00	0.00	0.00	Req per kernel
Misses - Non Duplicated	3116.00	0.00	49856.00	Req per kernel
Misses- Duplicated	0.00	0.00	0.00	Req per kernel
Cache Hit Rate	0.00	0.00	0.00	Pct
Read Req (Total)	3611.88	0.00	31488.00	Req per kernel
Atomic Req	0.00	0.00	0.00	Req per kernel
Read Req (1 DWord)	2508.00	0.00	26752.00	Req per kernel
Read Req (2 DWord)	0.00	0.00	0.00	Req per kernel
Read Req (4 DWord)	448.00	0.00	4736.00	Req per kernel
Read Req (8 DWord)	0.00	0.00	0.00	Req per kernel
Read Req (16 DWord)	655.88	0.00	6846.00	Req per kernel

14.3 Scalar L1D Cache - L2 Interface

Metric	Avg	Min	Max	Unit
sL1D-L2 BW	587976.00	0.00	3792000.00	Bytes per kernel
Read Req	3584.00	0.00	37888.00	Req per kernel
Write Req	5603.13	0.00	59250.00	Req per kernel
Atomic Req	0.00	0.00	0.00	Req per kernel
Stall Cycles	43.38	0.00	594.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

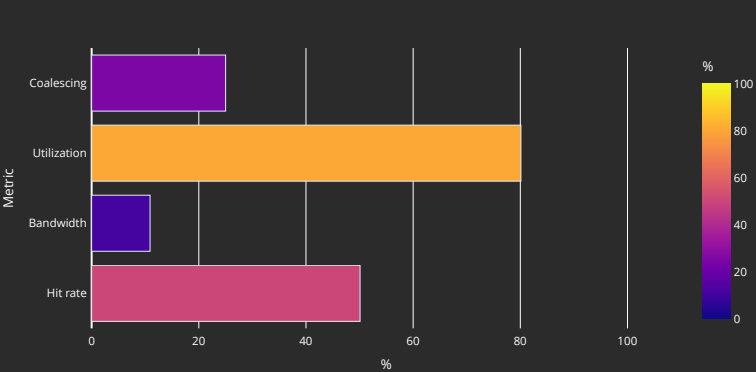
Metric	Avg	Min	Max	Unit
Address Processing Unit Busy	38.16	34.82	41.55	Pct
Address Stall	11.79	9.32	14.66	Pct
Data Stall	0.73	0.60	0.77	Pct
Data-Processor → Address Stall	0.00	0.00	0.00	Pct
Sequencer → TA Address Stall				Cycles per kernel
Sequencer → TA Command Stall				Cycles per kernel
Sequencer → TA Data Stall				Cycles per kernel
Total Instructions	1080081371.00	1080033280.00	1080802736.00	Instructions per kernel
Global/Generic Instructions	1077936128.00	1077936128.00	1077936128.00	Instructions per kernel
Global/Generic Read Instructions	1073741824.00	1073741824.00	1073741824.00	Instructions per kernel
Global/Generic Read Instructions for LDS				Instructions per kernel
Global/Generic Write Instructions	4194304.00	4194304.00	4194304.00	Instructions per kernel
Global/Generic Atomic Instructions	0.00	0.00	0.00	Instructions per kernel
Spill/Stack Instructions	2191870.00	2097152.00	2931680.00	Instructions per kernel
Spill/Stack Read Instructions	2146915.00	2097152.00	2533424.00	Instructions per kernel
Spill/Stack Read Instructions for LDS				Instructions per kernel
Spill/Stack Write Instructions	20064.00	0.00	321024.00	Instructions per kernel
Spill/Stack Atomic Instructions	0.00	0.00	0.00	Instructions per kernel
Spill/Stack Total Cycles	33997056.00	33554432.00	37445632.00	Cycles per kernel
Spill/Stack Coalesced Read	195260.00	0.00	1684416.00	Cycles per kernel
Spill/Stack Coalesced Write	83030.00	0.00	1328480.00	Cycles per kernel

15.2 Data-Return Path

Metric	Avg	Min	Max	Unit
Data-Return Busy	73.21	68.73	78.38	Pct
Cache RAM → Data-Return Stall	50.77	46.21	62.98	Pct
Workgroup manager → Data-Return Stall	0.00	0.00	0.00	Pct
Coalescable Instructions	88126.00	0.00	767968.00	Instructions per kernel
Read Instructions	1075842112.00	1075440720.00	1076608432.00	Instructions per kernel
Write Instructions	4239259.00	4194304.00	4592560.00	Instructions per kernel
Atomic Instructions	0.00	0.00	0.00	Instructions per kernel
Write Ack Instructions				Instructions per kernel

16. Vector L1 Data Cache

16.1 Speed-of-Light



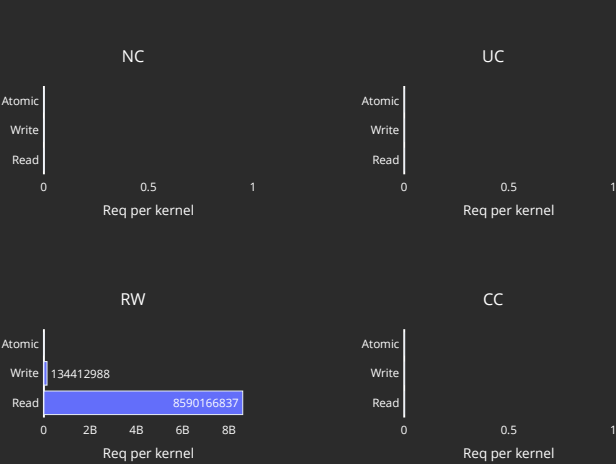
16.2 L1D Cache Stalls (%)

Metric	Min	Q1	Median	Q3	Max
Stalled on L2 Data	43.94	46.13	48.22	49.80	61.33
Stalled on L2 Req	0.00	0.01	0.02	0.02	0.11
Stalled on Address					
Stalled on Data					
Stalled on Latency FIFO					
Stalled on Request FIFO					
Stalled on Read Return					
Tag RAM Stall (Read)	0.00	0.00	0.00	0.00	0.00
Tag RAM Stall (Write)	0.17	0.22	0.24	0.26	0.26
Tag RAM Stall (Atomic)	0.00	0.00	0.00	0.00	0.00

16.3 L1D Cache Accesses

Metric	Avg	Min	Max	Unit
Total Req	69123874816.00	69122129920.00	69137285120.00	Req per kernel
Read Req	68854501888.00	68853694464.00	68860854272.00	Req per kernel
Write Req	269368560.00	268435456.00	276360960.00	Req per kernel
Atomic Req	0.00	0.00	0.00	Req per kernel
Cache BW	2237705950720.00	2237677961216.00	2238125793280.00	Bytes per kernel
Cache Hit Rate	50.09	50.08	50.11	Pct
Cache Accesses	17482077740.00	17481859072.00	17485357760.00	Req per kernel
Cache Hits	8757517077.63	8755785413.00	8761073810.00	Req per kernel
Invalidations	577.50	304.00	2704.00	Req per kernel
L1-L2 BW	1108142351152.00	1108118339584.00	1108232936832.00	Bytes per kernel
Tag RAM 0 Req				Req per kernel
Tag RAM 1 Req				Req per kernel
Tag RAM 2 Req				Req per kernel
Tag RAM 3 Req				Req per kernel
L1-L2 Read	8590163574.38	8590065664.00	8590908662.00	Req per kernel
L1-L2 Write	134397088.00	134217728.00	136007680.00	Req per kernel
L1-L2 Atomic	0.00	0.00	0.00	Req per kernel
L1 Access Latency				Cycles
L1-L2 Read Latency				Cycles
L1-L2 Write Latency				Cycles

16.4 L1D - L2 Transactions



16.5 L1D Addr Translation

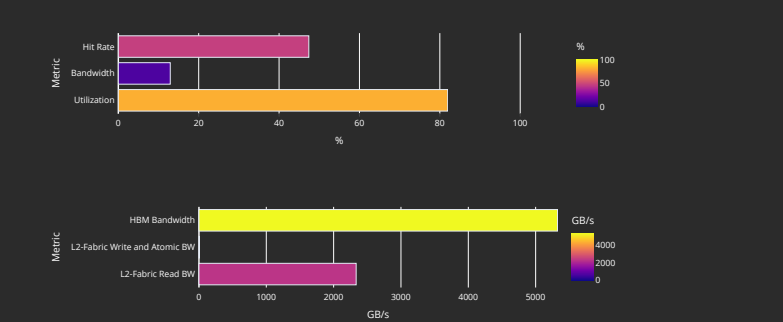
Metric	Avg	Min	Max	Units
Req	17482077740.00	17481859072.00	17485357760.00	Req per kernel
Inflight Req				Req per kernel
Hit Ratio	99.59	99.57	99.61	Pct
Hits	17410821107.75	17410385920.00	17414169111.00	Req per kernel
Translation Misses	67475702.56	67475456.00	67477472.00	Req per kernel
Misses under Translation Miss				Req per kernel
Permission Misses	0.00	0.00	0.00	Req per kernel

16.6 L1D Addr Translation Stalls

Metric	Avg	Min	Max	Units
Cache Full Stall				Cycles per kernel
Cache Miss Stall				Cycles per kernel
Serialization Stall				Cycles per kernel
Thrashing Stall				Cycles per kernel
Latency FIFO Stall				Cycles per kernel
Resident Page Full Stall				Cycles per kernel
UTCL2 Stall				Cycles per kernel

17. L2 Cache

17.1 Speed-of-Light



17.2 L2 - Fabric Transactions

Metric	Avg	Min	Max	Unit
Read BW	586771569868.00	566283176576.00	625857132288.00	Bytes per kernel
HBM Read Traffic	97.17	90.20	100.31	Pct
Remote Read Traffic	2.85	0.00	9.80	Pct
Uncached Read Traffic	0.00	0.00	0.00	Pct
Write and Atomic BW	2155561480.00	2147483648.00	2283521376.00	Bytes per kernel
HBM Write and Atomic Traffic	100.18	95.15	105.53	Pct
Remote Write and Atomic Traffic	0.41	0.00	4.85	Pct
Atomic Traffic	0.00	0.00	0.00	Pct
Uncached Write and Atomic Traffic	0.01	0.00	0.15	Pct
Read Latency	613.30	479.13	1097.34	Cycles
Write and Atomic Latency	230.09	156.83	508.20	Cycles
Atomic Latency				Cycles
Read Stall				Pct
Write Stall				Pct

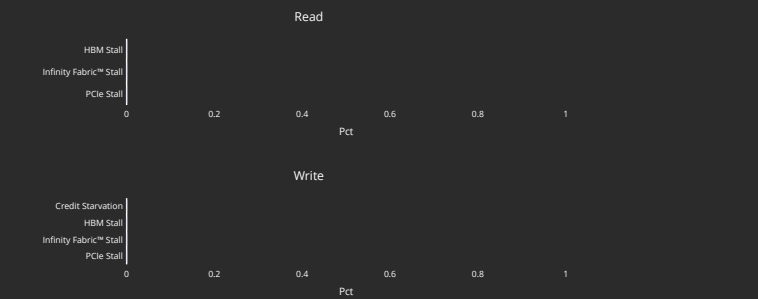
17.3 L2 Cache Accesses

Metric	Avg	Min	Max	Unit
Bandwidth	1116751209112.00	1116710137472.00	1117061963392.00	Bytes per kernel
Read Bandwidth				Bytes per kernel
Write Bandwidth				Bytes per kernel
Atomic Bandwidth				Bytes per kernel
Req	8724618821.19	8724297949.00	8727046589.00	Req per kernel
Read Req	8590192356.88	8590080251.00	8591051325.00	Req per kernel
Write Req	134413798.50	134217728.00	136068480.00	Req per kernel
Atomic Req	0.00	0.00	0.00	Req per kernel
Streaming Req	279168.00	0.00	2462720.00	Req per kernel
Bypassss Req				Req per kernel
Probe Req	3625.00	0.00	58000.00	Req per kernel
Input Buffer Req				Req per kernel
Cache Hit	47.40	44.36	49.10	Pct
Hits	4140062296.13	3812029704.00	4284023540.00	Hits per kernel
Misses	4594228349.94	4440870189.00	4939754585.00	Misses per kernel
Writeback	16833399.38	16777328.00	17554745.00	Cachelines per kernel
Writeback (Internal)	16826535.88	16732160.00	17612709.00	Cachelines per kernel
Writeback (vL1D Req)	43948.88	27774.00	73591.00	Cachelines per kernel
Evict (Internal)	4615895192.25	4440592380.00	4964886116.00	Cachelines per kernel
Evict (vL1D Req)	0.00	0.00	0.00	Cachelines per kernel
NC Req	0.00	0.00	0.00	Req per kernel
UC Req	1308.50	0.00	17855.00	Req per kernel
CC Req	0.00	0.00	0.00	Req per kernel
RW Req	8724616509.00	8724297933.00	8727029049.00	Req per kernel

17.4 L2 Cache Stalls

Metric	Avg	Min	Max	Unit
Stalled on Latency FIFO				Cycles per kernel
Stalled on Write Data FIFO				Cycles per kernel
Input Buffer Stalled on L2				Cycles per kernel

17.5 L2 - Fabric Interface Stalls



17.6 L2 - Fabric Detailed Transaction Breakdown

Metric	Avg	Min	Max	Unit
Read (32B)	0.00	0.00	0.00	Req per kernel
Read (64B)	30396942.31	~268773176.00	391814215.00	Req per kernel
Read (128B)	4568954418.44	4424096001.00	4865144388.00	Req per kernel
Read (Uncached)	4618.88	0.00	35936.00	Req per kernel
HBM Read	4462405225.75	4424077666.00	4544793344.00	Req per kernel
Remote Read	137826091.38	0.00	485047654.00	Req per kernel
Read Bandwidth - PCIe				Bytes per kernel
Read Bandwidth - Infinity Fabric™				Bytes per kernel
Read Bandwidth - HBM				Bytes per kernel
Write and Atomic (32B)	35358.88	~1721987.00	1711205.00	Req per kernel
Write and Atomic (Uncached)	3875.00	0.00	54400.00	Req per kernel
Write and Atomic (64B)	33662968.69	33554432.00	35291015.00	Req per kernel
HBM Write and Atomic	33753553.44	33554432.00	35425454.00	Req per kernel
Remote Write and Atomic	142533.00	0.00	1711141.00	Req per kernel
Write Bandwidth - PCIe				Bytes per kernel
Write Bandwidth - Infinity Fabric™				Bytes per kernel
Write Bandwidth - HBM				Bytes per kernel
Atomic	0.00	0.00	0.00	Req per kernel
Atomic - HBM				Req per kernel
Atomic Bandwidth - PCIe				Bytes per kernel
Atomic Bandwidth - Infinity Fabric™				Bytes per kernel
Atomic Bandwidth - HBM				Bytes per kernel